Every enumeration refines to a bijection

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openlogic:sets-functions-relations/size-of-sets:local-snapshot-20261013; CC BY 4.0

Editorial note. The counted statement is the author proposition labelled prop:enum-bij (912-916); the proof span 918-935 is the author proof environment that immediately follows it.

Editorial scope. One original result of the Open Logic Project unit sets-functions-relations/size-of-sets, counted once: every surjection from the positive integers onto A refines to a bijection from a finite initial segment or the positive integers onto A. Only the counted proposition and its own author proof are retained; the source is reproduced without mathematical changes and keeps the CC BY 4.0 licence and the author attribution.

Allowed standard prerequisites (editorial).

- Surjection and injection. A function is surjective when every element of the codomain is a value; it is injective when distinct arguments have distinct values; a bijection is both.. The constructed g must be shown both surjective and injective (975-978).
- Definition by recursion. A function whose value at i+1 is computed from the preceding values is well defined by recursion.. g(1) = f(1) and g(i+1) = the first value of f not already among g(1), ..., g(i) (963-968).

Proposition 0.1. *If $f\colon \mathbb{Z}^+ \to A$ is surjective (i.e., an enumeration of A), there is bijection $g\colon Z \to A$ where Z is either $\mathbb{Z}^+$ or $\{1, \dots, n\}$ for some $n \in \mathbb{Z}^+$.*

Proof. We define the function g recursively: Let $g(1) = f(1)$. If $g(i)$ has already been defined, let $g(i+1)$ be the first value of $f(1)$, $f(2)$, … not already among $g(1)$, …, $g(i)$, if there is one. If A has just n elements, then $g(1)$, …, $g(n)$ are all defined, and so we have defined a function $g\colon \{1, \dots, n\} \to A$. If A has infinitely many elements, then for any i there must be element of A in the enumeration $f(1)$, $f(2)$, …, which is not already among $g(1)$, …, $g(i)$. In this case we have defined a function $g\colon \mathbb{Z}^+ \to A$.

The function g is surjective, since any element of A is among $f(1)$, $f(2)$, … (since f is surjective) and so will eventually be a value of $g(i)$ for some i. It is also injective, since if there were $j < i$ such that $g(j) = g(i)$, then $g(i)$ would already be among $g(1)$, …, $g(i-1)$, contrary to how we defined g. □